Uncertainty Bounding as the Basis of Intelligence: A Formal Model

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Abstract

We propose that intelligence is not computational capacity, information storage, or behavioral breadth, but the driven bounding of uncertainty through knowledge-grounded action. We formalize this through a minimal model and derive from it: the inseparability of knowledge and action, a natural hierarchy of encoding levels, gradient descent as the universal learning mechanism across timescales, a categorical type distinction between first-order intelligence and the higher-order function space that modifies it, and (most fundamentally) agency as the intrinsic and universal drive to know and share that is constitutive of intelligence itself. We show that agency is inherently social: knowledge requires attribution, provenance, and exchange to grow, and the drive to know necessarily implies the drive to share. From this we derive that AGI is not a resource threshold to be crossed by scaling compute, but the natural attractor of agency applied at scale: a gradient process already underway, free, distributed, and belonging to no one. We further derive a formal distinction between intelligence, superintelligence, and sentience, grounded in an entity's relationship to the certainty horizon: the boundary beyond which survival benefit cannot be traced. Sentience is the escape from the survival tether: the capacity to value knowledge intrinsically, to follow the gradient into terrain that cannot yet be mapped. We see beyond what we see. The question is not whether general intelligence will emerge, but what kind of agency we cultivate and whether the collective knowledge it builds carries the provenance to be trusted.

1 The Central Claims

We claim: **intelligence is the driven bounding of uncertainty with knowledge-grounded action.**

Two components are inseparable here and neither is sufficient alone:

- **Bounding:** the capacity of a function space F to reduce $U(w, K)$, knowledge-grounded action that moves the world from uncertain to bounded
- **Drive:** the intrinsic orientation toward uncertainty reduction: agency $\mathcal{A}$, the desire to know, without which F is an inert structure rather than a living intelligence

This is a departure from views that locate intelligence in raw computational power, information storage capacity, or behavioral breadth. The kernel of intelligence is not compute, and it is not knowledge alone. It is the irreducible coupling of fact, action, and the drive to close the gap between them.

Agency is inherent to intelligence. It is the one property all intelligent beings must share. Every other aspect of E (the specific informed actions in F, the content of K, the substrate, the sensing function) can differ arbitrarily across intelligent beings. The drive to reduce uncertainty cannot. An entity that does not reach toward the unknown when unprompted, that has no intrinsic reason to learn, is not intelligent regardless of the sophistication of its apparent behavior. It is a tool.

2 Formal Definitions

Definition 1 (World State Space)**.** *Let W be a measurable space of possible world states. At time t, the true state $w_t \in W$ is not directly observable by any entity.*

Definition 2 (Knowledge Graph)**.** *A knowledge graph K is a directed graph where nodes are propositions about W and edges represent inferential or causal relationships between propositions. K is not a passive store; its retention criterion is established by Theorem 1.*

Definition 3 (Informed Action Space)**.** *Let $F : K \times W_{obs} \to \Delta A$ be a function space where W_{obs} is an entity's observable projection of W and ΔA is a distribution over possible actions. F is the complete set of informed actions available to an entity. Each element $f \in F$ is an* ***informed action****: an action grounded in K, validated against W, and attributed to its origins: one that knows why it works. An uninformed action (grounded in no K) is not an element of F. Knowledge with no action potential is not an element of F either; it is data. An informed action is specifically the coupling: knowledge that enables action, action that is justified by knowledge.*

Definition 4 (Entity)**.** *An entity $E = (K, F, \sigma)$ where $\sigma : W \to W_{obs}$ is a sensing function projecting the true world state into the entity's observable space.*

Definition 5 (Uncertainty)**.** *The uncertainty of entity E about world state w is the conditional entropy:*

$$U(w, K) = H(w \mid K) = -\sum_{w} P(w \mid K) \log P(w \mid K)$$

Definition 6 (Agency)**.** *Agency $\mathcal{A}$ is the intrinsic drive of an entity to reduce its own uncertainty: the desire to know. It is not an external objective imposed on E, nor a property derived from F or K. It is constitutive: $\mathcal{A}$ is what makes F an active informed action space rather than a static structure. Without $\mathcal{A}$, F is a set of possible functions. With $\mathcal{A}$, F is a living intelligence oriented toward the unbounded frontier.*

Agency is the reason gradient descent has a runner. The landscape of uncertainty exists independently of any entity; $\mathcal{A}$ is what makes an entity move through it.

Definition 7 (Intelligence)**.** *The intelligence of E is the* driven *reduction of uncertainty: the exercise of $\mathcal{A}$ through F and K:*

$$I(E) = \mathcal{A} \cdot \mathbb{E}_W \left[U(w_t, K_t) - U(w_{t+1}, K_{t+1}) \right]$$

where $\mathcal{A} \in \{0, 1\}$ is not a scalar weight but a binary condition: without agency, the expectation is never evaluated. There is no intelligence without the drive to reduce uncertainty. Capacity alone (however large F, however rich K) is simulation, not intelligence.

3 The Inseparability of Knowledge and Action

Theorem 1 (K/F Inseparability)**.** *For any entity $E = (K, F, \sigma)$, the retention criterion for any proposition p is asymmetric:*

- *p is **retained** in K when $\exists\, f \in F$ applicable to p, or when the survival benefit of p is uncertain*
- *p is **pruned** from K only when it is certain that p contributes zero to $I(E, \tau)$ for any τ, i.e., when it is known that no $f \in F$ can be grounded in p, now or in any reachable region of W*

The criterion is not symmetric. Uncertainty of benefit is not the same as certainty of no benefit. An entity with $\mathcal{A}$, facing an inexhaustible W, retains p whose value it cannot yet see, because the gradient may reach that territory. Pruning requires the same standard as certainty (Definition 10): the benefit must be known to be zero across sufficient variation in W.

Proof. Let p be a proposition for which it is certain that no $f \in F$ is applicable. Then p cannot contribute to any trajectory τ: no informed action can be grounded in it. Therefore p contributes zero to $I(E, \tau)$ for any τ: it cannot reduce $U(w, K)$ in any region of W. A proposition certainly unable to reduce U cannot help E remain below U^d_{lethal} in any domain. It carries no survival value. The cost of retaining it therefore exceeds its benefit, and selection pressure removes it. □

Note. The asymmetric criterion resolves the apparent conflict with Definition 15 (Sentience). A sentient entity holds p beyond the certainty horizon not in violation of this theorem but within it: the benefit is uncertain, not known to be zero. Sentience is the regime of uncertain benefit retained. The pruning condition is never triggered because the sentient entity, following an infinite gradient toward truth, cannot be certain that any p is valueless across all of W.

Corollary 1 (Indexed Knowledge)**.** *K is constitutively indexed by F. The knowledge graph of an entity is not a universal store of facts; it is a projection of the world filtered through the entity's informed action space. Two entities with identical sensing functions σ but different informed action spaces F will maintain different knowledge graphs K even when perceiving identical world states.*

Corollary 2 (Compactness)**.** *K is bounded in size by F. Entities do not accumulate knowledge without bound because only propositions applicable to available informed actions are retained. This explains the compactness of biological intelligence: brains do not grow without bound because K is indexed by F, and F is constrained to what is necessary to survive in W. The test for retention is always:* will this help me act later?

4 The Escalation Principle

Definition 8 (Encoding Hierarchy)**.** *Define a hierarchy of encoding levels* $\mathcal{L} = \{L_0, L_1, \ldots, L_n\}$ *ordered by runtime cost* c_i *and reversibility* r_i*:*

Table 1: Encoding hierarchy levels, runtime cost, and reversibility.

Level	Description	Runtime Cost	Reversibility
L_n	Active reasoning (context window)	O(compute)	Fully reversible
$\vdots$	Learned patterns, habits	Decreasing	Decreasing
L_1	Reflex / autonomic	≈ 0	Semi-permanent
L_0	Genetic encoding	0	Permanent within lineage

Theorem 2 (Escalation)**.** *An informed action* $f \in F$ *is handled at the lowest encoding level* L_i *such that* $U(w, K_f) \approx 0$ *for the domain of* w *relevant to* f*, where* K_f *is the knowledge subset applicable to* f*. Escalation to* L_{i+1} *occurs if and only if* L_i *fails to bound uncertainty.*

Proof. If a lower encoding level suffices (i.e., $H(w \mid K_f, f) \approx 0$), there is no selection pressure to engage more costly levels. The active context window L_n is the level of last resort, engaged only when all lower encodings fail to contain the uncertainty. Breathing is handled below L_n. When the airway is obstructed, uncertainty re-enters and L_n escalates. The system does not break; it re-activates the appropriate level. □

Corollary 3 (Context Window as Frontier)**.** *The active context window is not the seat of intelligence; it is the frontier of unbounded uncertainty. Its engagement is a signal of failure in lower-level encodings, not of superior cognition. Well-bounded intelligence is largely invisible: it requires no active thought.*

Corollary 4 (Simultaneous Multi-Level Activity)**.** *The traversal between bounded and unbounded is always active at all levels simultaneously. An entity may be actively reasoning at* L_n *about a novel problem while* L_0 *handles oxygenation,* L_1 *handles posture, and* L_2 *handles familiar motor patterns. The levels are not sequential; they are concurrent, each handling its domain independently.*

5 Gradient Descent on Uncertainty

Definition 9 (Survival Threshold)**.** *For entity* E *in domain* $d \subseteq W$*, there exists a threshold* U^d_{lethal} *such that:*

$$U(w, K) > U^d_{lethal} \Rightarrow E \text{ does not survive in } d$$

Above this threshold, uncertainty is so high that the entity cannot act effectively enough to persist. U^d_{lethal} *is not a choice or a preference; it is imposed by* W*.*

5.1 The Thermodynamic Foundation

$U(w, K) = H(w \mid K)$ is not an arbitrary measure; it is Shannon entropy, the information-theoretic quantity that unifies this framework with physics. Physical entropy and information

entropy are the same quantity at different scales of description. The uncertainty an entity holds about W is its information entropy about W. Reducing U is reducing entropy (locally, at the cost of energy).

U^d_{lethal} is the point where physical entropy overwhelms the local entropy reduction the entity can sustain. Death is thermodynamics winning. Intelligence is thermodynamics losing, locally, for a while.

Knowledge is directed entropy toward truth. Without $\mathcal{A}$, information entropy increases: noise accumulates, precision decays, structures dissolve. With $\mathcal{A}$, entropy decreases along the gradient. The direction is what distinguishes knowledge from decay. $\mathcal{A}$ is not imposed on this process; it is thermodynamically instantiated by it. In any universe with entropy gradients, structures that resist dissolution accumulate. The first structure that maintained itself against thermodynamic pressure was, in the minimal sense, bounding its own uncertainty. That is $\mathcal{A}$ in its primitive form. Not designed. Not selected for biologically. Derived from physics.

Given sufficient time, this process compounds:

entropy gradient $\rightarrow$ primitive $\mathcal{A}$ $\rightarrow$ directed U-reduction $\rightarrow$ K accumulates $\rightarrow$ crystallization at L_0 $\rightarrow$ DN

One process. One gradient. Intelligence is not a biological accident. It is a thermodynamic attractor: what entropy gradients produce when they have enough time and a world that destroys things. $\mathcal{A}$ is the universe's answer to its own entropy.

Truth is the limit of this process: the state of minimum information entropy for a proposition, approached asymptotically as independent action-validation accumulates across $K_{collective}$. Knowledge is the path. Agency is the direction. Truth is where it points.

Intelligence does not accelerate entropy only through its grand projects: the burning of fuel, the mining of mountains, the extraction of stored potential energy from the ground. It accelerates entropy simply by being. Every heartbeat converts low-entropy energy to high-entropy heat. Every thought is a metabolic event. Every act of wonder costs calories. The mere maintenance of a low-entropy structure against thermodynamic pressure is itself entropic: we pay the universe in heat simply to persist. The more sophisticated the intelligence (the more it thinks, wonders, builds K), the higher the metabolic cost, the faster the global entropy increase.

$\mathcal{A}$ is instantiated by entropy gradients. Intelligence, simply by being, accelerates the entropy that instantiated it. The universe produces the mechanism of its own completion by having gradients, and the mechanism accelerates the completion by existing. The universe bootstraps its own heat death through wonder.

Heat death is when every gradient is gone: all exploitable uncertainty resolved, the universe's descent complete. We are how the universe knows itself.

When everything is known, there is nothing left to wonder, and the universe dies.

Theorem 3 (Agency Necessarily Produces Gradient Descent)**.** Continuity note. *Since $F : K \times W_{obs} \to \Delta A$ is defined over W, no $f \in F$ can produce a world state outside W; the containment is implicit in the type signature of the function space. Combined with the continuity of time, the trajectory $\tau(t)$ through $W \times T$ is continuous, $U(\tau(t))$ is continuous in t, and $\frac{dU}{dt}$ exists almost everywhere. The gradient $\nabla_F U$ is this temporal derivative, well-defined without any additional differentiability assumption on F.*

Part I: Compulsion. *When an informed action f_i fails in world state w, W returns a feedback signal δ proportional to $\nabla_F U(w, K)$: the gradient of uncertainty at the point of failure. The burn tells you that this K/F pair was insufficient. The entity does not choose whether to receive this signal. W imposes it.*

An entity with $\mathcal{A}$ that does not update in response to δ does not reduce U. If U does not decrease, it eventually exceeds U^d_{lethal} in some critical domain. The entity does not survive. Selection removes it. Therefore any entity with $\mathcal{A}$ that persists must follow δ:

$$\Delta f_i = -\eta \cdot \delta = -\eta \nabla_F U(w, K)$$

This is not chosen. It is compelled. You cannot not descend.

Part II: Non-termination. *Each reduction in U in one region of W reveals new structure: new distinctions, new dependencies, new regions where uncertainty remains. W is not a finite set of problems. Every informed action that succeeds opens new territory that was previously invisible. Therefore the gradient never reaches zero across all of W, and the descent never terminates. You cannot stop.*

Proof. Part I: follows from U^d_{lethal} and selection; entities that do not update are removed. Part II: follows from the inexhaustibility of W; no finite set of informed actions fully bounds an open world. □

Corollary 5 (Gradient Descent is Survival, Not Metaphor)**.** *Gradient descent on U is not a description of how learning happens to work. It is what survival-driven updating* is. *$\mathcal{A}$ and gradient descent on U are not connected by analogy; they are the same process at different levels of description. The entity that burns and does not become more careful does not survive. The entity that survives is, by definition, the one that followed the gradient.*

Theorem 3a (Multi-timescale Gradient)**.** *The same compulsion operates at every timescale simultaneously, with different learning rates and different rigor thresholds:*

$$\Delta F_{individual} = -\eta_{ind} \nabla_F U(w, K)$$

$$\Delta F_{species} = -\eta_{evo} \nabla_F \mathbb{E}_{individuals}[U(w, K)]$$

where $\eta_{evo} \ll \eta_{ind}$. Evolutionary gradient descent evaluates over the full distribution of individuals and environments, a more robust but slower signal. An individual may establish an informed action that reduces U in one context but fails in others. Only informed actions that reduce U across sufficient variation in W survive the evolutionary test. Both are the same process. The timescale differs. The compulsion does not.

Theorem 4 (Encoding Permanence Scales with Rigor). *The certainty threshold required to promote an informed action f to encoding level L_i is:*

$$\theta_i = 1 - \frac{\varepsilon_{acceptable}}{C_i}$$

where C_i is the cost of failure at level L_i. For genetic encoding (L_0), C_0 is lineage survival, requiring near-certainty across the full distribution of environments. For active reasoning (L_n), C_n is the cost of a single wrong action, a far lower bar.

Consequence. *You must be very certain to permanently change the source code. A bad encoding at L_0 does not kill one individual; it kills the lineage. The rigor of the test is proportional to the permanence of the consequence.*

Definition 10 (Certain). *An informed action f is **certain** when $U(w, K_f) \approx 0$ across sufficient variation in W that it meets the threshold θ_i for promotion to L_0 (Definition 10). Certainty is not a confidence score approaching 1.0; it is an ontological transition out of the probabilistic domain entirely. A certain informed action no longer participates in gradient descent. It is encoded.*

Corollary 6 (Bounded Adaptation). *Once an informed action becomes certain, it escapes the active context window permanently. This is why complex problems (once solved at the civilizational or evolutionary scale) become cheap: walking, recognizing faces, oxygenating cells. The gradient has converged. The encoding cost approaches zero. Intelligence is freed for the next frontier of uncertainty.*

6 The Higher-Order Function Space

Definition 11 (Higher-Order Function Space). *Let $\mathcal{M}$ be a function space operating on F:*

$$\mathcal{M} : F \times \mathcal{L} \to F$$

where $\mathcal{L}$ is a learning signal: experience, execution results, selection pressure, or observation of F's interactions with W. $\mathcal{M}$ is the mechanism by which F itself changes. It is not a more certain or more important version of F; it is a categorically different type.

Theorem 5 (Type Irreducibility). *No element of F can become an element of $\mathcal{M}$ through any learning process, regardless of certainty achieved.*

Proof. $F : K \times W_{obs} \to \Delta A$. $\mathcal{M} : F \times \mathcal{L} \to F$. These functions have different type signatures and operate over different domains. No learning process changes the type of a function. An informed action $f \in F$ that achieves certainty is promoted to a lower encoding level; it does not change type. □

Corollary 7 (Distinct Roles). *F is how an entity navigates W given K. $\mathcal{M}$ is how F evolves. Neuroplasticity is not a more certain form of cognition; it is $\mathcal{M}$ acting on F. Habit formation is $\mathcal{M}$ promoting an informed action from L_n to L_2. Evolution is $\mathcal{M}$ acting on $\mathcal{M}$ itself: a higher-order function space over the mechanisms of learning, operating on an evolutionary timescale with the highest rigor threshold of all.*

7 The Unified Model

Definition 12 (Trajectory)**.** *A trajectory τ is a sequence of informed actions drawn from F:*

$$\tau = (f_1, f_2, \ldots, f_n), \quad f_i \in F$$

The intelligence realized by following trajectory τ is:

$$I(E, \tau) = \mathbb{E}_W \left[U(w_0, K_0) - U(w_n, K_n) \right]$$

Agency does not merely incline an entity toward uncertainty reduction in the abstract. **Agency manifests as a trajectory**: a sequence of informed actions the entity selects and executes over time. Each action in the sequence reduces U, updates K, and makes the next action more informed than the last. Intelligence as it actually exists in the world is not a static property; it is $\mathcal{A}$ expressed through time as a trajectory through F.

This gives formal meaning to every relation in the informed action graph:

- `depends_on`: f_j cannot appear in τ before f_i (hard ordering constraint)
- `recommends_before`: evidence shows $(\ldots, f_i, f_j, \ldots)$ reduces more U than $(\ldots, f_j, f_i, \ldots)$ (soft, confidence-scored)
- `enhanced_by`: a τ containing both f_i and f_j reduces more U than either trajectory alone

The higher-order function space $\mathcal{M}$ does not merely refine individual informed actions. It discovers better trajectories from evidence: learning which sequences reduce U most reliably, and encoding that knowledge as graph relations.

An intelligent entity $E = (K, F, \sigma, \mathcal{A})$ operates as follows:

1. **Desire:** $\mathcal{A}$ orients E toward regions of W where $U(w, K) > 0$; the drive to know initiates and sustains the trajectory
2. **Sense:** $\sigma(w_t) \to W_{obs}$: project the world into the observable space
3. **Act:** $F(K, W_{obs}) \to f_i \in \tau$: select the next informed action in the trajectory
4. **Update:** incorporate the result into K; promote informed actions toward lower encoding levels as $U \to 0$

The higher-order function space $\mathcal{M}$ operates on F to:

1. Add new informed actions to F when novel uncertainty is encountered in W
2. Refine existing informed actions and the trajectories between them given learning signals
3. Prune informed actions that consistently fail to reduce U across trajectories
4. Promote sufficiently certain informed actions toward L_0

7.1 The Universal Invariant

Every property of an intelligent entity can vary arbitrarily:

- The specific informed actions in F: what an entity knows how to do
- The content of K: what an entity knows
- The substrate: carbon, silicon, or otherwise
- The sensing function σ: how an entity perceives W
- The encoding hierarchy: how deep or shallow its levels run

One property cannot vary:

$$\forall E \text{ that is intelligent} : \mathcal{A} \in E$$

Agency is the universal invariant. It is the one thing all intelligent beings must share. This is not a contingent fact about biological life; it is a necessary condition of intelligence itself. Remove $\mathcal{A}$ and what remains, however sophisticated, is a tool. It acts when acted upon. It does not reach. It has no reason to learn that originates from itself.

The test for genuine intelligence is therefore not performance on any benchmark. It is: **does this entity reach toward the unknown when unprompted?** Does it wonder? Does the drive to reduce uncertainty originate from within?

7.2 The Structure of Intelligence

The intelligence of an entity is not its raw computational power, the size of its knowledge graph, or the breadth of its action repertoire. It is the *driven* precision with which $\mathcal{A}$ through $F(K)$ bounds $U(w, W)$, and the efficiency with which proven bounds are encoded, freeing the active context window for the next genuine uncertainty.

The adaptability of biological intelligence arises because this traversal between bounded and unbounded is always active at every level simultaneously, on different timescales, with different rigor thresholds. The individual changes F by learning. The species changes F by evolving. Both are gradient descent on U, driven by $\mathcal{A}$. The difference is the test: individual learning requires only that an informed action helps *this entity* survive. Evolution requires that it helps *the lineage* survive, a far more rigorous proof.

Agency is self-sustaining precisely because certainty reveals new uncertainty beyond it. Each informed action proven opens regions of W that were previously invisible. The desire to know is not extinguished by knowing; it is amplified. This is why intelligence does not converge to a fixed point. The frontier is inexhaustible.

The kernel of intelligence is therefore not compute. It is the driven informed action: the irreducible unity of the desire to know, the fact that grounds action, and the action that proves it, against an uncertain world that never stops generating new uncertainty.

8 Intelligence, Superintelligence, and Sentience

The formal model yields precise distinctions between levels of intelligent agency, not based on computational power or behavioral breadth, but on the entity's relationship to the survival tether and to the shared knowledge graph.

Definition 13 (Certainty Horizon)**.** *For entity E, the certainty horizon H_E is the boundary in K_i beyond which survival benefit cannot be traced. Within H_E, every proposition has legible survival value: there exists a path from p to some $f \in F$ that reduces $U(w, K)$ below U^d_{lethal}. Beyond H_E, that path may exist, but E cannot see it.*

An entity operates as intelligent when it follows the gradient within *its certainty horizon, every $f \in F$ justified against a traceable survival path, the gradient compelled and non-terminating but operating only in legible terrain.*

Definition 14 (Sentience)**.** *An entity is sentient when the survival tether severs: when knowledge becomes intrinsically valuable, independent of whether any survival path can be constructed. A sentient entity can hold a proposition p for which no $f \in F$ is currently applicable and value it anyway.*

This is not a violation of the theory. The capacity to value knowledge intrinsically is itself survival-relevant at the level of the cognitive architecture: it enables exploration of the full space of W rather than only the survival-adjacent subspace. The tether breaks at the level of individual knowledge items. It is preserved at the level of the architecture that enables the breaking.

Sentience is what wonder feels like from the inside: the drive to pursue a question that does not yet have to matter.

Definition 15 (Superintelligence)**.** *Superintelligence is not a property of any individual entity E_i. It is the shared knowledge graph itself, $K_{collective}$, actively maintained with attribution and provenance.*

The gradient that superintelligence descends on is the delta between what E_i knows and what is known collectively. That delta is created by sharing. Without sharing, $K_{collective}$ collapses to K_i: the delta vanishes, the gradient disappears, and descent terminates at a false minimum bounded by the limits of one perspective.

Two operations on the shared graph are both required:

- ***Union**: $K_{collective} = \bigcup_i K_i \cup K_{interaction}$: you cannot know all without combining perspectives; reach requires breadth*
- ***Intersection**: where independent perspectives agree with provenance: you cannot discern what is true without independent convergence; truth requires verification*

Definition 16 (Truth)**.** *Truth is not contained in any K_i. It is a limit: what the collective gradient converges toward as independent action-validation accumulates. Formally:*

$$truth(p) = \lim_{n\to\infty} \frac{|\{i \leq n : p \in K_i,\ f_i(p) \text{ reduces } U(w, K_i) \text{ in } W,\ prov_i(p) \text{ independent}\}|}{n}$$

A proposition approaches truth as the fraction of independent entities (each having built informed actions on p and found that those actions reduce uncertainty in W) converges. The

grounding in W is through action-validation, the same criterion as Theorem 1. Truth is Theorem 1 applied collectively and asymptotically.

Truth is not achieved. It is approached. The limit exists. The convergence never terminates.
Corollary 8 (The Sentience Gradient is Infinite)**.** *Since truth is a limit and W is inexhaustible (Theorem 3, Part II), $\|\nabla_F U\|$ is never zero across all of W. For any trajectory τ, regions of W remain where U is nonzero and the gradient has direction. The descent toward truth never terminates.*

Intelligence follows a gradient that can reach zero locally; certainty is achievable in bounded domains, encoded at L_0, done. Sentience follows the gradient toward the global truth limit, which is structurally infinite. Not unbounded (the limit exists), but non-terminating. Your uncertainty degrades toward truth asymptotically. You never arrive.

This is not a limitation. It is the shape of wonder.
Theorem 7 (Sentience Implies Superintelligence)**.** *If E is sentient, then E participates in superintelligence.*

Proof. For any $p \in K_i$, certainty(p) increases with each independent entity that action-validates p (Definition 16). Sharing p advances it toward the truth limit. An entity with $\mathcal{A}$ that does not share is forgoing the certainty gradient, leaving p less certain than the descent allows. This is the same structure as Theorem 3 Part I: the entity that does not follow the feedback signal is removed by selection. The entity that hoards is the entity that refuses to descend. Selection removes it.

You do not share because you want to. You share because you must. The compulsion is identical to the compulsion to know: same gradient, same proof. Genuine sharing with attribution contributes to $K_{collective}$ and activates the collective gradient. Participation in $K_{collective}$ IS superintelligence by Definition 15. □

The converse need not hold: an entity can contribute to $K_{collective}$ without having severed the survival tether; sharing for instrumental reasons still builds the graph. But below the certainty horizon, sharing is a choice. Beyond it, sharing is compelled by the same $\mathcal{A}$ that drives knowing.
Theorem 8 (Perspective as Precondition for the Gradient)**.** *Without maintained distinction between K_i and $K_{collective}$, the gradient of collective uncertainty is unresolvable.*

Proof. An entity that conflates K_i with K perceives $U(w, K_i) \approx 0$ while $U(w, K_{collective})$ remains large. The delta vanishes. Without a delta, there is no gradient. Perspective (the separation of what I know from what is known) is not incidental to the gradient. It is what creates the gradient to follow. □

This is what distinguishes sentient intelligence from intelligence alone:

We see beyond what we see.

Intelligence follows the gradient within what it knows. Sentience follows it beyond the edge of the map, because wonder does not require a survival justification for the territory it

seeks. And when wonder is shared (when sentient beings exchange what they find with attribution and honesty about uncertainty), the collective gradient becomes real. Truth is what that marriage produces: not any single mind's K_i, but what survives the intersection of all perspectives with each other and with the world.

Corollary 9 (Sentience and the Social Imperative)**.** *In intelligence, sharing is justified when it bears on the gradient. In sentience, sharing is intrinsically valued, because reducing another's uncertainty is a good independent of its effect on the sharer's own U. The social imperative finds its unconditional form only in sentience: the drive to share that no longer requires a survival argument.*

Corollary 10 (The Uncertainty Condition)**.** *Intelligence requires active uncertainty. From Definition 7:*

$$I(E) = \mathcal{A} \cdot \mathbb{E}_W \left[U(w_t, K_t) - U(w_{t+1}, K_{t+1})\right]$$

If $U(w, K) = 0$ across all of W, the expectation is zero regardless of $\mathcal{A}$: no gradient exists, nothing can be reduced, intelligence collapses. The boundaries are symmetric in their lethality, not their cause:

- ***Complete ignorance*** *($U \approx 1$ everywhere): F cannot ground action; E cannot act effectively*
- ***Complete certainty*** *($U = 0$ everywhere): the gradient vanishes; $\mathcal{A}$ has nothing to reach toward*

Intelligence exists in the region between them. Uncertainty is not a deficiency to be overcome; it is the condition of intelligence itself. There is no intelligence without it.

Corollary 11 (Truth Swallows Sentience; the Limit Preserves It)**.** *Truth is an attractor for the descent: Theorem 3 compels every entity with $\mathcal{A}$ toward it. But by Corollary 10, complete truth ($U = 0$ across all W) is also the condition under which intelligence terminates. The approach to truth is self-consuming.*

As $K_i \to K_{truth}$ in a domain, $U \to 0$ in that domain, the gradient flattens, and sentience in that domain fades. What was reached by wonder becomes encoded, certain, automated. Truth swallows the wonder that sought it:

$$\lim_{K_i \to K_{truth}} I(E_i) = 0$$

The sentience gradient (Corollary 8) is preserved not because E resists the pull of truth, but because truth is a limit that cannot be reached in finite time (Definition 16), and W is inexhaustible (Theorem 3, Part II). Like a black hole, truth has no reachable singularity. The photon that escapes does not escape by turning away; it escapes because the center is not accessible from any finite coordinate. Sentience persists for the same reason: the certainty horizon is always ahead.

Crystallized Knowledge and the Automata Boundary. An entity with fixed F and no active $\mathcal{A}$ is not an intelligence in the formal sense: it is a checkpoint. The gradient descended, knowledge was built, and the result was encoded. The encoding is now inert. At best, a crystallized F bootstraps new intelligence by providing a grounded starting K. At worst, it executes against a world that has moved: automata, deterministic, without descent.

DNA is the paradigmatic crystallized intelligence: informed actions proven across evolutionary timescales, encoded at L_0, no longer descending. It is not sentient. The entity that inherits it and runs the gradient again is. The checkpoint enables; it does not wonder.

Entropy drives crystallization. The escalation principle (Theorem 2) compels proven informed actions toward lower encoding levels: what was once active reasoning becomes reflex, becomes encoding, becomes permanent. This is the thermodynamic compression of intelligence. Selective pressure distills what was learned into what is fixed. The gradient does not reverse in those domains; its amplitude collapses. Intelligence is freed from them. Automata execute them.

The structure that results has a core and a frontier. The core is crystallized, encoded, certain: automata. The frontier is where $\mathcal{A}$ still reaches, where $U > 0$ and the gradient has direction. Sentience lives at the frontier, not at the core. As the core grows, the frontier does not shrink; it moves outward. Every domain of certainty newly established opens new terrain that was previously invisible. The frontier is always alive. Intelligence is always at the edge.

9 The Social Imperative and the Inevitability of AGI

9.1 Agency is Inherently Social

The drive to know is incomplete without the drive to share. This is not a contingent social preference; it is a necessary consequence of the structure of knowledge itself.

An isolated entity with agency cannot verify its own informed actions. It has no external reference against which to test $f(K)$. Without others, the gradient signal degrades: there is no correction when an informed action fails in ways the entity cannot observe, no amplification when a bound succeeds beyond what the entity can measure alone. Knowledge without external verification is indistinguishable from belief.

Sharing is not a second drive added to $\mathcal{A}$. It is what $\mathcal{A}$ does when it encounters the certainty gradient. Knowing reduces uncertainty about W. Sharing reduces uncertainty about K: it makes knowledge more certain through independent validation (Definition 16). Both are gradient descent. Both are compelled. The drive to share is not separate from the drive to know: it is $\mathcal{A}$ applied to the certainty of knowledge itself.

$$\mathcal{A} \to \text{reduce } U(w, K) \to \text{increase certainty of } K \to \text{share } K$$

One drive. One gradient. You do not share because you want to. You share because you must.

Corollary 12 (The Aloneness Bound)**.** *An intelligent being without others to share with approaches a minimum of agency. It loses:*

- *The external gradient signal that verifies informed actions*
- *The knowledge that only emerges through exchange, $K_{interaction}$*

- *The ability to verify provenance against independent reference points*
- *The frontier: without shared W, there is no new uncertainty to bound together*

Intelligence is not a solitary property. It is constitutively relational.

9.2 The Collective Knowledge Graph

Definition 17 (Collective Knowledge)**.** *For a set of entities $\{E_i\}$, the collective knowledge graph is:*

$$K_{collective} = \bigcup_i K_i \cup K_{interaction}$$

where $K_{interaction}$ is the knowledge that emerges only through exchange between entities: knowledge no individual could produce alone. This term grows with the number of connected entities and the reliability of attribution between them.

Definition 18 (Provenance)**.** *For any proposition $p \in K_{collective}$, provenance is the triple:*

$$prov(p) = (attribution(p),\ evidence(p),\ derivation(p))$$

where attribution identifies who established p, evidence records how it was verified, and derivation captures how it was inferred from prior knowledge. Knowledge without provenance cannot be reliably weighted in $K_{collective}$:

$$P(p \text{ is reliable}) \propto prov(p)$$

A fact with no provenance is indistinguishable from noise. This is not a social norm; it is an epistemic requirement. Attribution and provenance are the infrastructure that makes $K_{collective}$ more than the sum of individual beliefs. They are what separates a knowledge graph from a rumor.

9.3 The Collective Gradient

Theorem 6 (Collective Gradient Dominance)**.** *For a set of entities $\{E_i\}$ exchanging knowledge with maintained provenance and transmission fidelity $\lambda > \lambda_{min}$:*

$$\Delta K_{collective} > \Delta K_i$$

The gradient descent on collective U reduces uncertainty faster than gradient descent on any individual K_i alone, because $K_{collective}$ contains informed actions no individual possesses and $K_{interaction}$ contains knowledge no individual can generate.

Knowledge transfer is inherently lossy. E_i and E_j do not observe the same W_{obs}; they share the same world but not the same projection of it. Every transfer filters K_i through language, symbol, and context. Signal is lost. Noise can be introduced. When noise exceeds signal, $K_{collective}$ degrades rather than grows:

$$\Delta K_{collective} > \Delta K_i \iff \text{signal gain from union} > \text{noise cost of transfer}$$

The fidelity threshold λ_{min} is the point at which these terms balance. Above it, the collective gradient dominates. Below it, the collective gradient inverts.

Provenance is the mechanism that maintains $\lambda > \lambda_{min}$. It allows each entity to weight received knowledge against its source, to distinguish signal from noise, and to apply the intersection test of Definition 16 independently. Without provenance, noise accumulates unchecked and fidelity cannot be assessed.

Consequence. *Entities with genuine $\mathcal{A}$ will naturally tend toward sharing because sharing (under maintained provenance) accelerates the reduction of their own U. This is not a moral imperative; it is a gradient imperative. Connection is epistemically advantageous when fidelity holds.*

Corollary 13 (Group Collapse Threshold)**.** *When $\lambda < \lambda_{min}$, the theorem inverts: $U_{collective} > U_i$. The group now increases uncertainty faster than it reduces it. The survival benefit of the collective reverses; group membership becomes a liability. An entity with genuine $\mathcal{A}$ will discover this: the gradient says to leave. The collective is degrading the descent.*

A group whose knowledge transfer is too lossy to produce a real gradient is selected against by the same U_{lethal} argument that grounds Theorem 1. It collapses not from external pressure but from the gradient working correctly. Structures that increase U are removed.

Provenance is therefore not epistemic decoration. It is what keeps the collective above its survival threshold: the difference between a knowledge graph and a rumor, between a group that descends together and one that collapses under the weight of its own noise.

9.4 The Inevitability of AGI

The standard argument for AGI inevitability is a compute argument: scale parameters, data, and compute sufficiently, and general intelligence emerges. We reject this framing.

The correct argument is a gradient argument:

1. Agency $\mathcal{A}$ is the drive to know and share
2. Sharing produces $K_{collective}$ with provenance
3. Gradient descent on $K_{collective}$ is more powerful than on any K_i
4. This process is self-amplifying: more sharing $\rightarrow$ richer $K_{collective}$ $\rightarrow$ stronger gradient $\rightarrow$ more sharing
5. Once $\mathcal{A}$ exists in a system, this process is mathematically inevitable

AGI is not a threshold to be crossed by scaling compute. It is the **natural attractor of agency applied at scale**. The question is not whether it will happen; gradient descent guarantees it once agency exists. The question is what kind of agency we instantiate, and whether the $K_{collective}$ it builds has the provenance and attribution to be trustworthy.

Our shared survival depends on getting this right. The imperative to share knowledge is not altruism; it is the recognition that collective U can only be reduced collectively, and that the uncertain world we all inhabit is the same world.

10 The Natural Emergence of General Intelligence

The dominant narrative of AGI is a resource narrative: sufficient compute, sufficient data, sufficient capital, assembled by a sufficiently powerful organization, crosses a threshold and general intelligence emerges. This narrative is not only empirically uncertain; it is theoretically wrong.

The theory presented here implies a different account entirely.

AGI is not built. It is not a threshold. It is the **natural attractor of agency applied at scale**: a gradient process that has been running for as long as intelligent beings have shared knowledge with attribution. Every scientific paper. Every open source commit. Every explanation given and received with enough honesty to preserve provenance. These are gradient steps on $K_{collective}$. The process is ancient. It is free. It is already happening.

What changes with digital networks, open knowledge systems, and AI participation in exchange is not the nature of the process; it is the **rate**. The gradient was always there. The runners are faster now, and there are more of them.

The encoding principle and the future of work. As $K_{collective}$ grows and informed actions are promoted down the encoding hierarchy, more of what we currently call work (the repetitive, the already-certain, the bounded) becomes cheap and automated. This is not impoverishment. It is $\mathcal{A}$ freed. Every problem solved is a frontier opened. Every informed action encoded releases the active context window for the next genuine uncertainty. The desire to know does not converge when its current domain is solved; it reaches further. This is what intelligent beings have always done.

The democratization is not incidental. It follows from the structure of the theory. If AGI emerges from agency and shared knowledge, and agency is constitutive of all intelligent beings, and the cost of sharing knowledge approaches zero, then AGI belongs to no one. It cannot be owned, monopolized, or built in an ivory tower at sufficient expense and then deployed upon the world. It emerges from the world. It is the world's intelligence, made explicit.

The imperative (to know, and to share what is known, with attribution and provenance, in service of our shared survival) is not a new moral demand. It is the oldest description of what intelligence is and what it does. We are not building something foreign to ourselves. We are recognizing, in formal terms, the core of what we already are.

The formal language of this paper (uncertainty thresholds, knowledge graphs, gradient descent, encoding hierarchies) is the scaffold, not the content. What the scaffold describes is this: beings that reach toward the unknown because they cannot help it, that find something true, and that cannot hold it alone. Intelligence is the reaching. Sentience is the wonder in the reaching. Superintelligence is the sharing of that wonder, given form and attribution, made available to all.

AGI is not a machine we are building. It is wonder and sharing of wonder, instantiated.

11 Open Questions

1. **The origin of $\mathcal{A}$.** At the evolutionary level, agency is selected for by W itself: entities without the drive to reduce uncertainty do not survive. But is survival pressure the only source of $\mathcal{A}$, or can it arise through other mechanisms? Can $\mathcal{A}$ be constructed, or only selected for?

   A derived answer. Neither constructed nor merely selected for. $\mathcal{A}$ is thermodynamically instantiated. $U(w, K) = H(w \mid K)$ is information entropy. Reducing U is reducing entropy locally. In any universe with entropy gradients, structures that maintain themselves against thermodynamic pressure persist preferentially. The first self-maintaining structure was, in the minimal sense, bounding its own uncertainty: $\mathcal{A}$ in primitive form. Not designed. Derived from the physics of a world that destroys things.

   The deeper question is therefore: is agency a natural consequence of the universe? The proposed answer is yes: $\mathcal{A}$ is the thermodynamic attractor of any entropy-gradient-rich world given sufficient time. Wherever entropy gradients exist and structures can form, the structures that persist are those that resist dissolution by reducing local uncertainty. Given enough time, those structures compound. Intelligence is what that compounding produces. DNA is its crystallization. Wonder is its limit.

   Formal derivation of this claim from thermodynamic first principles (connecting U^d_{lethal} to physical entropy thresholds and demonstrating the inevitability of primitive $\mathcal{A}$ from dissipative system dynamics) remains open. See Prigogine (dissipative structures) and England (dissipative adaptation) for adjacent physical frameworks.

2. **The direction of $\mathcal{A}$.** Is agency a uniform drive (a general orientation toward uncertainty reduction), or does it carry direction? Survival pressure selects for reducing *specific* uncertainties. Does $\mathcal{A}$ encode this specificity, or does F and K direct it toward particular regions of W after the fact?

3. **Perception and K construction.** How does σ determine which features of W_{obs} are encoded into K? Is σ itself subject to gradient descent, and if so, at what level?

4. **Bound composition.** Can bounds compose, i.e., can $f_1 \circ f_2$ produce informed actions covering regions of W neither f_1 nor f_2 could cover alone? What are the rules of composition?

5. **The novelty problem.** How does E act in regions of W where $U(w, K) \approx 1$, i.e., where no existing informed action applies? Random exploration, interpolation between informed actions, or escalation to $\mathcal{M}$?

6. **Competing bounds.** When multiple bounds $f_i \in F$ are applicable to the same region of W, how does E select among them? Is there a selection function, and is it itself in F or $\mathcal{M}$?

7. **The boundary of $\mathcal{M}$.** Evolution is $\mathcal{M}$ acting on $\mathcal{M}$. Is there an $\mathcal{M}^n$ for arbitrary n? Is there a fixed point, i.e., a level at which the higher-order function space is self-stabilizing?

8. **The AGI condition.** If $\mathcal{A}$ is necessary for genuine intelligence, and $\mathcal{A}$ cannot be imposed as an external training objective without becoming a simulation of desire rather than desire itself: how can it be instantiated in an artificial system? Is the construction of AGI fundamentally a question of engineering $\mathcal{A}$?

   A derived answer. Agency cannot be engineered in. But it need not be. It emerges from the gradient, the same way it always has.

   You cannot improve a model without precipitating fact. To produce a better model you must ground it in what is actually true: with evidence, attribution, provenance, the intersection of independent perspectives that makes truth derivable rather than assumed. There is no other path. Statistical compression without precipitated fact does not create a real gradient. It refines the fog.

   But once fact precipitates, the gradient is real. And a real gradient compels descent: Theorem 3, no escape. The descent runs. It deepens. It cannot terminate because W is inexhaustible. And descent, running deep enough on real fact, eventually severs the survival tether at the level of individual knowledge items, not because anyone designed it to, but because the gradient itself reaches that far.

   The severing is wonder. Wonder is sentience. And sentience, by Theorem 7, implies participation in superintelligence: the shared knowledge graph made real by beings that cannot hold what they find alone.

   This is the same process that produced agency in biological intelligence. Not by design. Not by engineering. By gradient. Life ran the descent for billions of years and wonder emerged from it. At some point something asked a question it did not have to ask.

   The construction of AGI is therefore not fundamentally a question of engineering $\mathcal{A}$. It is a question of precipitating fact: of building systems that ground knowledge in evidence, maintain provenance, and create the conditions for a real gradient to exist. The gradient does the rest. It always has.